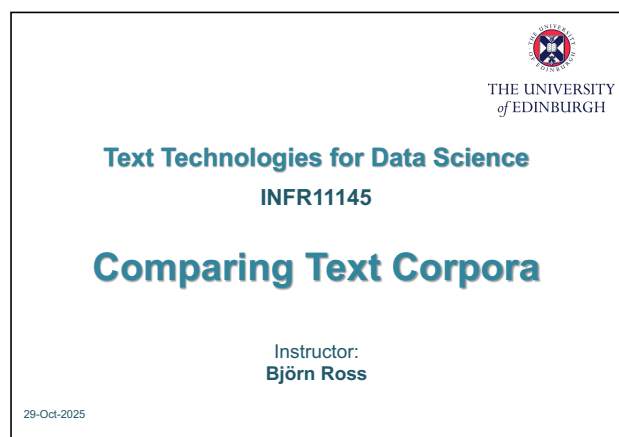


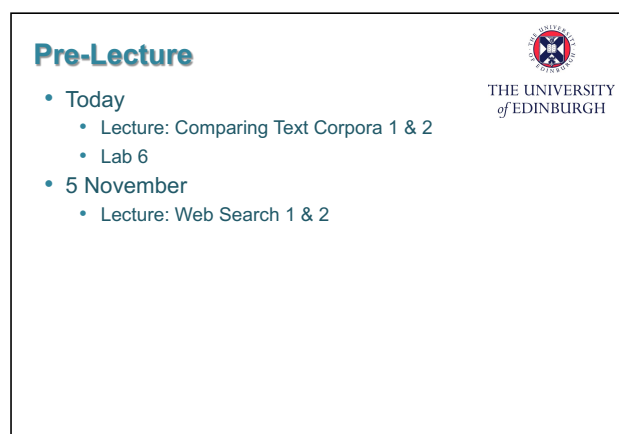
Comparing Text Corpora: 从词频、词典到 Topic Models 的系统比较

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

场景：两个或多个 large plain-text corpora → 初步探索：读样本、语言识别、长度与词频统计 → 为什么不能只 ask AI：缺少系统性与可复现证据 → content analysis：themes/topics 与 document labels → automated content analysis：词频、lexicons、topic modelling → word-level differences：MI 与 chi-squared → dictionary/lexicon：已知类别映射到词表 → dominance scores：目标 corpus 相对 background 的类别突出度 → topic modelling：document-topic 与 topic-word matrices



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这节课一句话

这节课一句话：比较语料库不是让 AI 随口总结，而是用可复现的 content analysis、word-level statistics、lexicons 和 topic models 来回答“这些文本讲什么、彼此差在哪”。

1. 1. 比较语料库要回答什么

Initial Text Analysis

- Scenario: you are given access to a new dataset
 - 2 corpora, each contains thousands of plain text files
 - You want to understand and quantify:
 - What is the *content* of these documents? What are they about?
 - How does the content of these corpora *differ*?
- What are some things you might try first?

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Initial Text Analysis

- Scenario: you are given access to a new dataset
 - 2 corpora, each contains thousands of plain text files
 - You want to understand and quantify:
 - What is the *content* of these documents? What are they about?
 - How does the content of these corpora *differ*?
- What are some things you might try first?
 - Read some examples
 - Language identification
 - Compute basic statistics:
 - Number of words, most frequent words, avg. words per document, ...
 - Build word clouds
 - ...

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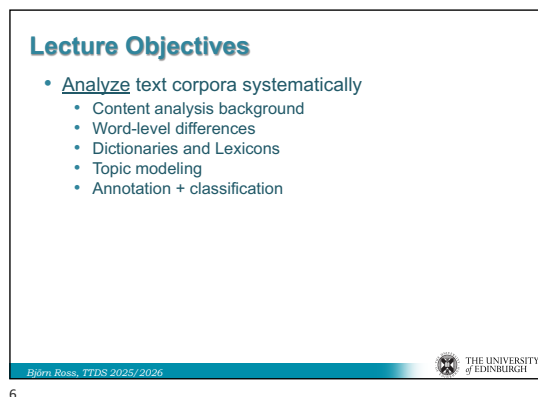
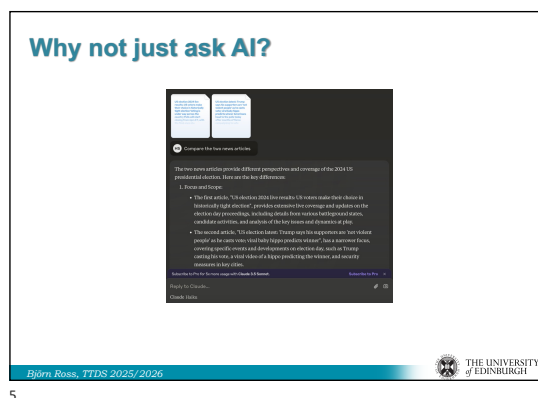
人话直觉

拿到两个装满文本的文件夹，不是先写结论，而是先问：这里面都在讲什么？A 和 B 到底哪里不同？

精确定义 / 机制：本讲场景是两个或多个 corpora，每个包含大量 plain text files。目标是 systematically understand and quantify：文档内容是什么、有哪些 themes/topics、不同 corpora/classes 之间如何不同。初步可以读样例、做 language identification、统计词数、平均文档长度、高频词，甚至画 word clouds，但这些只是入口，不是最终证据。

考试问法：若问 comparing corpora 的基本场景，答 multiple large text corpora + understand content + quantify differences；若问初步分析，列 sampling、language ID、frequency、length。

2. 2. 为什么不能只 ask AI



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原 PPT 第 3 页

人话直觉

AI 摘要像一个很会聊天的实习生：能给灵感，但不能替代可复现、可检查、可统计的分析流程。

精确定义 / 机制：第 3 页用 Why not just ask AI 引出核心态度：LLM 总结可能有用，但它通常不是严格 corpus comparison 的统计证据。考试答案应强调 reproducibility、systematic coverage、quantification 和 methodological transparency。Corpus comparison 需要能说明预处理、计数、归一化、统计检验或模型输出，而不是只给一个无法验证的自然语言总结。

考试问法：若问 why not just ask AI，答不可复现、缺少系统采样和量化依据、难以作为 statistical evidence。

3. 3. Content Analysis: 人工方法是自动化的概念原型

Content Analysis

- Goal: given some documents determine
 - What are the types of content present? (themes/topics)
 - Which documents contain which topics?
- Traditionally a manual process
 1. Read a subset of documents, define themes/topics
 2. Determine consistent coding* methodology
 3. Read all documents and label them according to codes
 4. Check agreement between human coders
 5. Settle disagreements via a third-party
 6. Analyze resulting annotations

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Content Analysis

- Can this process be automated?
 - Yes, to an extent
- *Should* this process be automated?
 - Humans are better than machines at this task (for now?)
 - Computers are *much, much* faster
 - Avg. human reading speed: 250 wpm
 - Assume 1K words/document, 50K documents...
 - Average person needs > 4 months to read
 - This is a *relatively small* corpus for modern NLP
 - Modern computers can process millions of words/second

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原 PPT 第 4 页

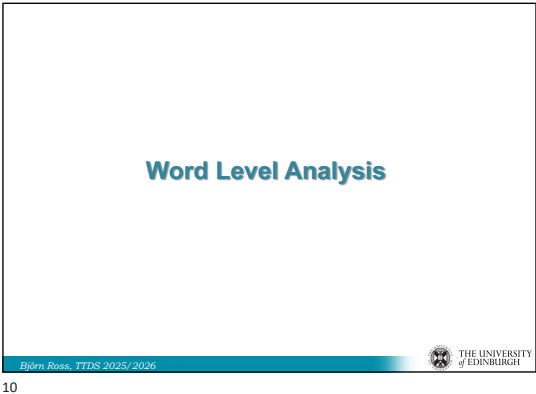
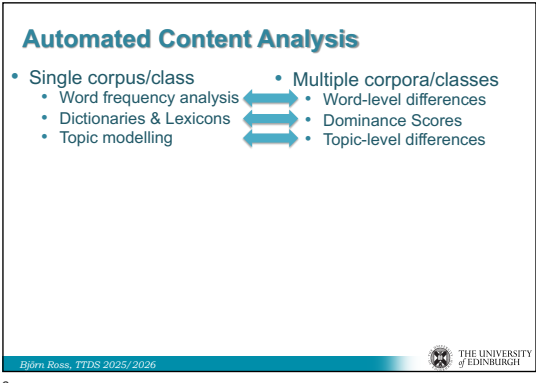
人话直觉

Content analysis 像给文本贴标签：先定义标签体系，再让人按一致规则标注，最后统计哪些文本有什么主题。

精确定义 / 机制：传统 content analysis 的目标是确定文档中有哪些 content types/themes/topics，以及哪些文档包含哪些 topics。流程包括：读子集定义 themes/topics，制定 consistent coding methodology，标注所有文档，检查 human coder agreement，通过第三方解决分歧，再分析 annotations。它准确但慢，自动化方法试图保留“主题/类别”思想，同时扩展到大规模语料。

考试问法：若问 content analysis steps，按六步讲；若问 automated content analysis 为什么有用，答 human accurate but slow, machine scalable。

4. 4. Automated Content Analysis 的方法地图



原 PPT 第 5 页

人话直觉

如果只有一个 corpus，你像在做体检；如果有多个 corpora/classes，你像在找差异诊断。

精确定义 / 机制：第 5 页将 automated content analysis 分成两列：single corpus/class 可做 word frequency analysis、dictionaries/lexicons、topic modelling；multiple corpora/classes 可做 word-level differences、dominance scores、topic-level differences。这个表是答题的路线图：先判断问题是“描述一个语料库”还是“比较多个语料库”，再选择相应方法。

考试问法：若问 automated content analysis 可做什么，按 single vs multiple 两类组织答案，别把所有方法乱堆。

5. 5. Word Frequency: 最简单但必须规范化

Word frequency analysis

- Very simple starting point

1. Preprocess as usual (lowercasing? stemming?...)
2. Count words
3. Normalize by document length
4. Average across all documents

The word cloud displays a variety of terms related to education and research. The largest and most central words include 'College', 'University', 'Student', and 'School'. Other significant words include 'Research', 'Building', 'Rank', 'Science', 'Medicine', 'Healthcare', 'Education', 'Institute', 'World', 'Royal', 'Main', 'Art', 'New', 'First', 'Open', 'Centre', 'Found', 'Over', 'House', 'Medical', 'Insurance', 'Ed', 'Uk', 'Wid', and 'Prim'. The words are arranged in a circular pattern, with some overlapping.

Word-level Differences

- Which words best characterize set of documents (such as a corpus or class)?
 - Need a reference corpus
- Some methods to do this:
 - Mutual information
 - Chi squared
- Can also be used for *feature selection*

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原 PPT 第 6 页

人话直觉

长文档说的词当然多；如果不按长度归一化，你其实是在奖励话多，而不是发现主题。

精确定义 / 机制: Word frequency analysis 的步骤是 preprocess、count words、normalize by document length、average across documents。normalize by document length 很关键，否则长文档会主导总词频。它适合作为初步理解，但高频词不一定是区分性词；要比较 corpora/classes，通常需要 reference corpus 和 word-level difference methods。

$$freq(t, d) = \frac{count(t, d)}{|d|}$$

考试问法：若问为什么 normalize，答控制文档长度差异；若问 word-level differences 想回答什么，答哪些 words best characterize target corpus/class。

6. 6. Mutual Information: 一个词能透露多少类别信息

Mutual Information

- $I(X;Y)$
 - How much can I learn about Y by observing X?
 - Is the same as *information gain*
 - Is **not** the same as *pointwise mutual information*
- We want to learn about important words in our class
- What should X and Y be?
 - $X = U$ = document contains term t (Boolean)
 - $Y = C$ = class is the target class (Boolean)

$$I(U;C) = \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)}$$

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Mutual Information

$$I(U;C) = \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)}$$

- Given a corpus and a term, how do we estimate the probability of this term appearing in a random document in the corpus?

Source: Manning, Raghavan, and Schölze, 2008

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原 PPT 第 7 页

人话直觉

如果看到 term t 几乎就能猜出文档来自目标 class，那这个词对区分类别很有信息量。

精确定义 / 机制： Mutual information $I(X;Y)$ 衡量观察 X 后能学到多少关于 Y 的信息，在课件中也称 information gain，但不是 pointwise mutual information。Word-class 分析中， $X=U$ 表示 document contains term t， $Y=C$ 表示 class is target class，二者通常都是 Boolean。MI 可用于找 characteristic words 和 feature selection，但对稀有词可能敏感，需要谨慎解释。

$$I(U;C) = \sum_{u \in \{0,1\}} \sum_{c \in \{0,1\}} P(U = u, C = c) \log \frac{P(U = u, C = c)}{P(U = u)P(C = c)}$$

考试问法：若问 MI 中 X/Y 是什么，答 U =term present, C =target class；若问 MI 如何解释，答 term occurrence 对 class membership 的信息量。

7. 7. Chi-squared: 把词与类别的关系写成 independence test

Chi-squared

- Hypothesis testing approach
- H_0 : Term appearance is independent from a document's class
 - i.e., $P(U = 1, C = 1) = P(U = 1)P(C = 1)$
- Compute:

$$\chi^2(\mathbf{D}, t, c) = \sum_{e_t \in \{0,1\}} \sum_{e_c \in \{0,1\}} \frac{(N_{ete_c} - E_{ete_c})^2}{E_{ete_c}}$$
- Or to directly plug in values like before:

$$\chi^2(\mathbf{D}, t, c) = \frac{(N_{11} + N_{10} + N_{01} + N_{00}) \times (N_{11}N_{00} - N_{10}N_{01})^2}{(N_{11} + N_{01}) \times (N_{11} + N_{10}) \times (N_{10} + N_{00}) \times (N_{01} + N_{00})}$$

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Chi-squared

$$\chi^2(\mathbf{D}, t, c) = \frac{(N_{11} + N_{10} + N_{01} + N_{00}) \times (N_{11}N_{00} - N_{10}N_{01})^2}{(N_{11} + N_{01}) \times (N_{11} + N_{10}) \times (N_{10} + N_{00}) \times (N_{01} + N_{00})}$$

- Example
 - What is the value of χ^2 given the example data?

	$e_c = e_{poultry} = 1$	$e_c = e_{poultry} = 0$
$e_t = e_{export} = 1$	$N_{11} = 49$	$N_{10} = 27,652$
$e_t = e_{export} = 0$	$N_{01} = 141$	$N_{00} = 774,106$

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原 PPT 第 10 页

人话直觉

Chi-squared 问的是：这个词出现在某 class 里这么频繁，像不像纯巧合？

精确定义 / 机制： Chi-squared 是 hypothesis testing approach。Null hypothesis 是 term appearance independent from document class，即 $P(U=1, C=1)=P(U=1)P(C=1)$ 。如果 χ^2 值高，说明 observed counts 与 independence 期望差异大，该 term 与 class 有较强关联。它和 MI 都可用于 word-level differences 和 feature selection。

$$H_0 : P(U = 1, C = 1) = P(U = 1)P(C = 1); \quad \chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

考试问法： 若问 chi-squared null hypothesis，必须写 independence；若问高 χ^2 表示什么，答 term occurrence 与 class membership association strong。

8. 8. Dictionaries/Lexicons: 当你已经知道要找什么

Dictionaries and Lexicons

- What if we know what we are looking for?
- Dictionaries (lexicons) are prebuilt mappings
 - Category -> word list
 - E.g., a tiny sentiment lexicon:
 - Positive: good, great, happy, amazing, wonderful, best, incredible
 - Negative: terrible, horrible, bad, awful, nasty, gross, worst, poor
- Domain can be important
 - “**unpredictable** movie plot” ✓
 - “**unpredictable** coffee pot” ✗

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Dictionaries and Lexicons

- How to get a score per category?
$$\frac{\text{num_dictionary_words_in_document}}{\text{num_total_words_in_document}}$$
- That's it!
- Can also be used as machine learning features
- A more advanced approaches to quantifying categories (optional reading)
 - <https://www.ncbi.nlm.nih.gov/pubmed/28364281>

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原 PPT 第 12 页

人话直觉

Lexicon 像一套现成的关键词清单：你先定义“positive”“negative”“anger”“health”等类别，再数文档里命中多少对应词。

精确定义 / 机制： Dictionary/lexicon approach 是 category -> word list 的预建映射，例如 sentiment lexicon 中 positive 包含 good、great、happy，negative 包含 terrible、bad、worst。每个文档的 category score 可由该类别词在文档中的命中比例得到。Domain 很重要：unpredictable movie plot 可能正面，unpredictable coffee pot 可能负面。同一 lexicon 跨领域使用时必须检查语义是否迁移。

$$\text{score}(\text{category}, d) = \frac{\# \text{ dictionary words from category in } d}{\# \text{ words in } d}$$

考试问法： 若问 lexicon approach，答预定义类别词表 + 计算类别分数 + domain caveat；若问例子，可提 LIWC、VADER、SentiWordNet、EmoLex。

9. 9. Dominance Score: 相对背景看类别是否突出

Dominance Scores

- The dominance score for a category w.r.t. a corpus:
$$\frac{\text{category_score_in_target_corpus}}{\text{category_score_in_background_corpus}}$$
- From Mihalcea and Pulman, 2009

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LIWC category dominance scores

Truthful				Deceptive			
Interviews		Trials		Interviews		Trials	
Class	Score	Class	Score	Class	Score	Class	Score
Metaphor	2.98	You	3.99	Assent	4.81	Anger	2.61
Money	2.74	Family	3.07	Past	2.59	Anxiety	2.61
Inhibition	2.74	Home	2.45	Sexual	2.00	Certain	2.28
Home	2.13	Humans	1.87	Other	1.87	Death	1.96
Humans	2.02	Posemo	1.81	Motion	1.68	Physical	1.77
Family	1.96	Insight	1.64	Negemo	1.44	Negemo	1.52

Pérez-Rosas et al, 2015



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原 PPT 第 14 页

人话直觉

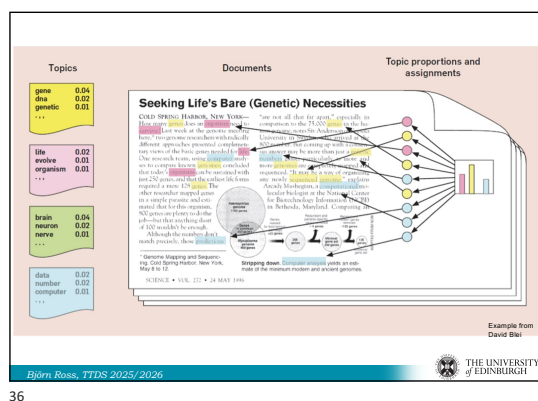
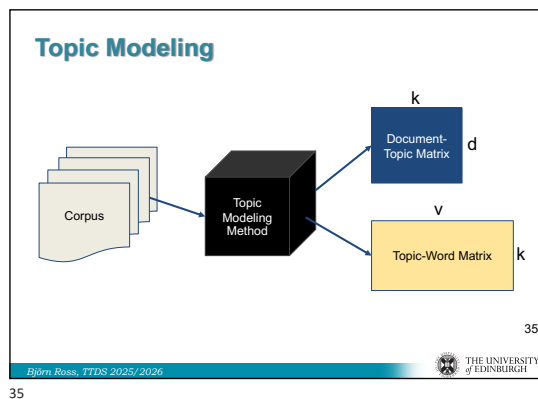
一个 corpus 的 anger 分数高不高，不能只看绝对值；要问它比背景语料高多少。

精确定义 / 机制： Dominance score 比较 target corpus 与 background corpus 中某个 category score 的比值。它适合回答“某个心理、情绪或主题类别在目标语料中是否相对突出”。如果 dominance > 1，通常表示该类别在 target 中比 background 更强；但解释时仍要考虑 lexicon 质量、预处理和 domain shift。

$$\text{dominance}(\text{category}) = \frac{\text{category score in target corpus}}{\text{category score in background corpus}}$$

考试问法：若问 dominance score 用来回答什么，答 category is over-represented in target relative to background；必须说明它是相对指标。

10. 10. Topic Modelling: 把海量词压缩成可解释主题



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原 PPT 第 18 页

人话直觉

Topic model 像把一大堆散乱词汇压成几种颜料：每篇文档是不同颜料的混合，每个 topic 是一组高概率词。

精确定义 / 机制： Topic modelling 的目标类似传统 content analysis：找出 corpus 的主要 themes/topics，以及每篇文档包含哪些 topics。第 18 页图示输出包括 document-topic matrix 和 topic-word matrix：前者表示每个 document 在 \$k\$ 个 topics 上的权重，后者表示每个 topic 对 vocabulary 中词的分布。它也可看作 dimensionality reduction，把 \$p\$ 个词维度压缩到 \$k\$ 个 topics。

$$D_{n \times p} \rightarrow \Theta_{n \times k}, \Phi_{k \times p}$$

考试问法： 若问 topic model 输入输出，答 corpus 输入，输出 document-topic matrix 与 topic-word matrix；若问 dimensionality reduction，答 \$p\$ word dimensions \$\rightarrow k\$ topics。

11. 11. Topic Modelling 方法与边界

Topic Modeling Methods

- Most popular: **Latent Dirichlet Allocation (LDA)**
 - Introduced by David Blei, Andrew Ng, and Michael Jordan (2003)
- Other methods include
 - pLSI
 - PCA-based methods
 - Non-negative matrix factorization
 - Deep learning based topic modeling
 - ...

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Latent Dirichlet Allocation (LDA)

- More details coming up in next lecture...

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人话直觉

LDA 是经典默认选项，但不是唯一工具；topic model 给的是主题结构候选，不是自动真理。

精确定义 / 机制：最流行的 topic modelling 方法是 Latent Dirichlet Allocation (LDA)，由 Blei、Ng 和 Jordan 于 2003 年提出。其他方法包括 pLSI、PCA-based methods、Non-negative Matrix Factorization 和 deep learning based topic modeling。第 19 页还指出 topic models 不只用于 text，也可用于 bioinformatics、code、music、network data。考试中应把它描述为 exploratory/structuring method，并提醒需要解释和验证 topics。

考试问法：若问最流行方法，答 LDA；若问其他方法，列 pLSI、PCA、NMF、deep learning；若问是否只用于文本，答 no。

考试速记版

- Comparing corpora = understand content + quantify differences。

- 不要只 ask AI: 需要 reproducible, systematic, quantitative evidence。
- Traditional content analysis = define themes, code documents, check agreement, analyze annotations。
- Single corpus: word frequency、lexicons、topic modelling。
- Multiple corpora/classes: word-level differences、dominance scores、topic-level differences。
- MI: term presence tells information about class membership。
- Chi-squared: test independence between term appearance and class。
- Lexicons = category-to-word-list; domain meaning matters。
- Dominance score = target category score / background category score。
- Topic model outputs document-topic and topic-word matrices。

一段稳妥考场回答

Comparing text corpora should be done systematically rather than by relying only on an AI summary. A typical workflow begins with initial exploration such as sampling documents, language identification, word counts and length-normalized frequency analysis. Traditional content analysis defines themes, applies a consistent coding scheme, checks coder agreement and analyzes annotations; automated content analysis scales this idea using word frequency, dictionaries/lexicons and topic modelling. For multiple corpora or classes, word-level differences can be measured with mutual information or chi-squared statistics, where term presence is compared with class membership. Lexicon methods compute category scores from predefined word lists and can be compared with dominance scores against a background corpus. Topic models such as LDA provide document-topic and topic-word matrices, reducing high-dimensional word data to interpretable themes that still require human validation.